

Title of Your Thesis

An optional subtitle

Firstname Lastname

Master's thesis

Master of Science in Economics

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Table of contents

Abstract	3
1 Introduction	4
2 Writing with Quarto	5
2.1 Citations	5
2.2 Equations	6
2.3 Figures	6
2.4 Tables	7
2.5 Cross-references	8
2.6 Workflow	9
3 Conclusion	10
References	11
Appendices	12
A Additional tables	12
Declaration of authorship	13

Abstract

State your research question, the data and method you use, and your main result in a single paragraph of about 150 to 250 words. Write it last, once you know what the thesis says.

1 Introduction

Replace this chapter with your own introduction. Within the first page or two, tell the reader what question you ask, why it matters, how you answer it, what you find, and what your work adds to what is already known.

End with a short roadmap. In this template, Chapter [2](#) is a worked example of the tools you need to write a thesis in Quarto: citations, equations, figures, tables and cross-references. Chapter [3](#) concludes, and Appendix [A](#) shows how an appendix is numbered. The last page holds the declaration of authorship you sign.

2 Writing with Quarto

This chapter is a worked example. Open `guide.qmd` next to the PDF to see how each element is written, copy what you need into your own chapters, then delete this one and remove it from `chapters: in _quarto.yml`. Writing text and code in one document goes back to what Knuth (1984) called literate programming: the analysis and its description can no longer drift apart.

The example asks an old question in empirical economics: do countries with more dependants save less? Under the life-cycle hypothesis (Modigliani 1966), people save during their working years and run their savings down in retirement, so countries with more young and old dependants should save a smaller share of their income. The data are R's built-in `LifeCycleSavings` set, 50 countries averaged over 1960 to 1970, which Belsley et al. (1980) took from a bachelor's thesis written at MIT.¹

Load the packages and prepare the data once, at the top of the chapter:

```
library(ggplot2)
library(modelsummary)

# Income in thousands, so its coefficient is readable
savings <- LifeCycleSavings
savings$dpi_k <- savings$dpi / 1000
```

2.1 Citations

References live in `references.bib`, a plain-text BibTeX file that a reference manager such as Zotero (with the Better BibTeX add-on) can keep up to date for you. Every entry has a key, and you cite it by writing the key after an `@`:

You write	You get
<code>@modigliani1966</code> argues	Modigliani (1966) argues
the data [<code>@belsley1980</code>]	the data (Belsley et al. 1980)
[see <code>@modigliani1966</code>]	(see Modigliani 1966)
[<code>@knuth84</code> ; <code>@r2024</code>]	(Knuth 1984; R Core Team 2024)

¹Footnotes like this one are written inline, right where they belong, and numbered automatically.

Add a page number with `[@modigliani1966, p. 12]`. The list of references at the end holds every source you cite and nothing else, in the Chicago author-date style unless you choose another with `csl: in_quarto.yml`. Cite your software too: `citation()` and `citation("modelsummary")` in R print ready-made entries.

2.2 Equations

Equations use LaTeX notation between double dollar signs. A label after the closing `$$` numbers the equation:

$$sr_i = \beta_0 + \beta_1 \text{pop15}_i + \beta_2 \text{pop75}_i + \beta_3 \text{dpi}_i + \beta_4 \text{ddpi}_i + \varepsilon_i \quad (2.1)$$

Here sr_i is the savings ratio of country i , aggregate personal saving as a percentage of disposable income; pop15 and pop75 are the shares of the population under 15 and over 75; dpi is real disposable income per head and ddpi its growth rate. Inline maths such as $\beta_1 < 0$ goes between single dollar signs.

2.3 Figures

Figures come from code chunks. The options at the top of a chunk, the lines starting with `#|`, give the figure a label that begins with `fig-` and a caption, and Figure 2.1 is then numbered and can be referenced anywhere.

```
ggplot(savings, aes(x = pop15, y = sr)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x,
              colour = "black") +
  labs(x = "Population under 15 (%)",
       y = "Savings ratio (%)") +
  theme_minimal()
```

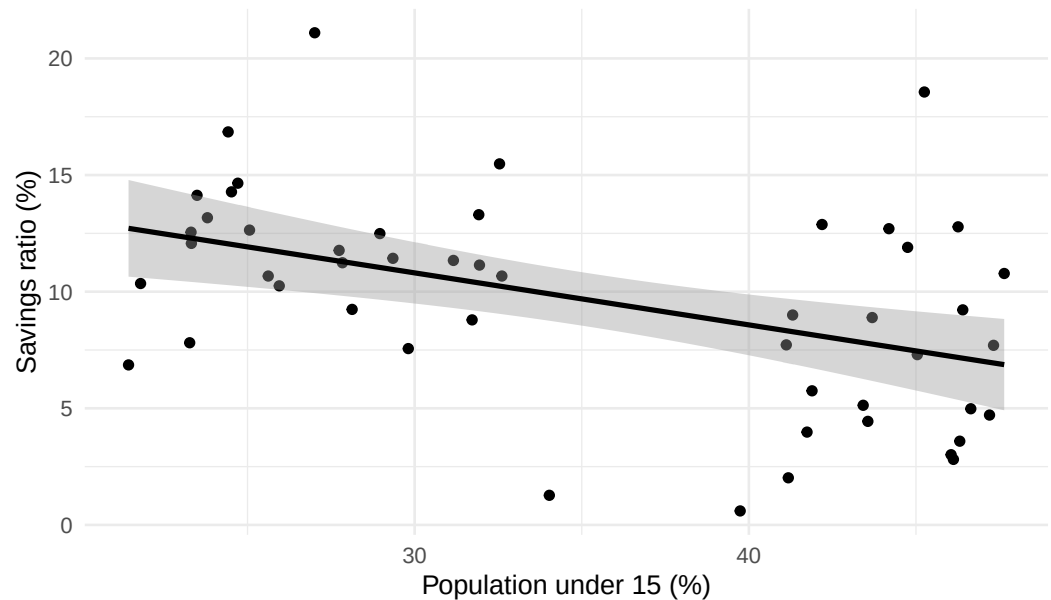


Figure 2.1: Savings ratio and the share of the population under 15, 50 countries, averages for 1960 to 1970

Code is shown here so that you can see how each result is made. To hide it in your thesis, add `#| echo: false` to a chunk, or `echo: false` under `execute:` in `_quarto.yml` for all chunks at once.

2.4 Tables

Tables work the same way, with a label that begins with `tbl-`. The `modelsummary` package (Arel-Bundock 2022) turns fitted models into regression tables that look right in both the PDF and the web version:

```
models <- list(
  "(1)" = lm(sr ~ pop15 + pop75, data = savings),
  "(2)" = lm(sr ~ pop15 + pop75 + dpi_k + ddpi,
             data = savings)
)

modelsummary(
  models,
  stars = c("*" = 0.1, "***" = 0.05, "****" = 0.01),
  coef_map = c(
    pop15 = "Population under 15 (%)",
    pop75 = "Population over 75 (%)",
    dpi_k = "Disposable income per head (thousands)",
    ddpi = "Growth of disposable income (%)",
    "(Intercept)" = "Constant"
  ),
)
```

```
gof_map = c("nobs", "r.squared")
)
```

Table 2.1: The savings ratio and the age structure of the population, OLS estimates of Equation 2.1

	(1)	(2)
Population under 15 (%)	−0.471*** (0.147)	−0.461*** (0.145)
Population over 75 (%)	−1.934* (1.041)	−1.691 (1.084)
Disposable income per head (thousands)		−0.337 (0.931)
Growth of disposable income (%)		0.410** (0.196)
Constant	30.628*** (7.409)	28.566*** (7.355)
Num.Obs.	50	50
R2	0.262	0.338

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

In column (2) of Table 2.1, a share of under-15s one percentage point higher goes with a savings ratio 0.46 points lower. That number is not typed in: a short piece of R code inside the sentence computes it, so it changes whenever the data or the model do. Standard errors are in parentheses; `vcov = "HC1"` in `modelsummary()` switches to heteroskedasticity-robust ones (this needs the `sandwich` package). The option `#|tbl-pos: "H"` keeps the table where it is written; without it, LaTeX may move a table to the top or bottom of a page. Descriptive statistics for all five variables are in Table A.1 in the appendix.

2.5 Cross-references

Anything with a label can be referenced with `@` and its label: writing `@fig-savings`, `@tbl-savings`, `@eq-lifecycle` and `@sec-citations` gives Figure 2.1, Table 2.1, Equation 2.1 and Section 2.1. Section labels such as `{#sec-citations}` go after a heading. The numbers update when you add, move or delete figures, tables and sections, so never type them by hand.

2.6 Workflow

A few habits keep a thesis reproducible from the first draft to the final PDF:

- **Render often.** `quarto preview` shows the web version and refreshes it whenever you save; `quarto render`, or Render Book in RStudio's Build pane, builds the PDF in `_book/`. Problems are easiest to fix while they are small.
- **Keep data and code in the project.** Put raw data in a `data/` folder, read it with relative paths such as `read.csv("data/survey.csv")`, and never edit it by hand. Each chapter runs in its own R session, so load the packages and data it needs at its top, as this one does.
- **Let the computer remember.** `freeze: auto` in `_quarto.yml` re-runs a chapter's code only when that chapter changes, and `renv` records the package versions you use in `renv.lock`; run `renv::snapshot()` after installing a new package.
- **Use Git.** Commit whenever you finish a piece of work, with a message that says what changed, and push to GitHub, so that every version of your thesis is backed up.

3 Conclusion

Replace this chapter with your conclusion. Summarise what you found, say what it means and where its limits lie, and suggest what future work could do next. New results do not belong here.

References

- Arel-Bundock, Vincent. 2022. “modelssummary: Data and Model Summaries in R.” *Journal of Statistical Software* 103 (1): 1–23. <https://doi.org/10.18637/jss.v103.i01>.
- Belsley, David A., Edwin Kuh, and Roy E. Welsch. 1980. *Regression Diagnostics: Identifying Influential Data and Sources of Collinearity*. Wiley.
- Knuth, Donald E. 1984. “Literate Programming.” *The Computer Journal* 27 (2): 97–111. <https://doi.org/10.1093/comjnl/27.2.97>.
- Modigliani, Franco. 1966. “The Life Cycle Hypothesis of Saving, the Demand for Wealth and the Supply of Capital.” *Social Research* 33 (2): 160–217.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.

Appendix A

Additional tables

Appendix chapters are listed under `appendices:` in `_quarto.yml`. They are lettered rather than numbered, so tables and figures in this one are numbered A.1, A.2 and so on. The code that makes Table A.1 is hidden with `# | echo: false`, as it usually would be in a thesis.

Table A.1: Descriptive statistics, 50 countries, averages for 1960 to 1970

	Mean	SD	Min	Max
Savings ratio (%)	9.67	4.48	0.60	21.10
Population under 15 (%)	35.09	9.15	21.44	47.64
Population over 75 (%)	2.29	1.29	0.56	4.70
Disposable income per head (thousands)	1.11	0.99	0.09	4.00
Growth of disposable income (%)	3.76	2.87	0.22	16.71

Declaration of authorship

Replace this text with the declaration your examination office requires, word for word. You will find the current wording on your examination office's forms (Prüfungsamt) or in your programme's examination regulations (Prüfungsordnung). It typically confirms that you wrote the thesis yourself, used only the sources and aids you name, and have not submitted it for another examination; many universities now also ask you to state whether and how you used AI tools.

Place, date

Signature